

Fig. 4: PCB pattern of the RGB colour code generator

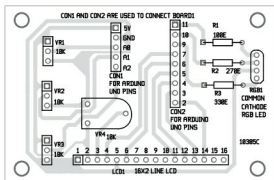


Fig. 5: Component layout of the PCB

the duty cycle of the PWM, thereby controlling the intensity of the colour. The RGB LED emits the light of a colour, which is controlled by the potentiometers (VR1 through VR3). The corresponding colour value is displayed in the first row as Rxxx Gxxx Bxxx, where xxx are the numbers for RGB colours, respectively. The second line displays the hex value of the colour as HEXxxxxxx. RGB colours with numbers are displayed on the LCD as shown in Fig. 3.

Resistor values (R1, R2 and R3) are critical. Choose these values after calibration. For calibration, connect each lead of the RGB LED in series with a 1-kilo-ohm resistor to 5V. Vary the potentiometer to low-resistance side until the RGB LED

emits a slightly whitish colour. Measure resistance from the arm of the potentiometer to the anode of the LED and note down the resistance value, say, 100-ohm. Using the same technique, measure the remaining two leads of the LED. These resistance values were found to be 270-ohm and 330-ohm, respectively, during testing.

Be careful while calibrating as there is a chance of burning the potentiometer when the arm moves to very low-resistance side due to more current passing through the arm.

Construction and testing

An actual-size, single-side PCB for the RGB colour code generator is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on a designed PCB. Connectors CON1 and CON2 in the PCB layout are used to connect Board1 (Arduino UNO board) through external connectors. Arduino board can be powered by an external 9V, 500mA adaptor or USB cable.

Software for the RGB generator is written in Arduino programming language. Arduino UNO is programmed using Arduino IDE. Select the correct board from Board->Tools menu in Arduino IDE, select COM port and upload the program (RGB.ino) through the standard USB port in computer.

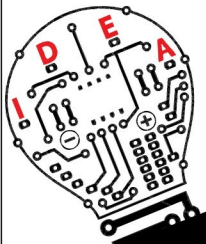
Note. For backlight, connect pin 15 of LCD1 to +5V via a 220-ohm resistor and pin 16 to GND. **EFY**



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